

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)

Full Stack development MERN Practical

Module - II

Practical –VIII
(Mini Project)

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Class:TYIT	Batch:2
Date of Assignment: 23/09/26	Date/Time of Submission: 23/09/26

Title: College Canteen Pre-Order System

Develop a web-based College Canteen Pre-Order System. It should allow students to:

1. View the available food items and their prices.
2. Enter student details such as name, roll number, and course.
3. Select a food item and quantity.
4. Place an order using a POST API.
5. Store the order details in MongoDB.
6. Retrieve the stored orders using a GET API.
7. Display the available food items and placed orders on the React page.

APIs to be implemented

- GET /food – Retrieve food items.
- POST /orders – Add a new food order.
- GET /orders – Retrieve placed orders.

Technologies

- React
- Axios
- Node.js
- Express.js
- MongoDB

Expected Output

The final application should contain:

- Food Item List
- Student Order Form
- Place Order button
- Order List/Table

Design the interface and decide the appropriate fields, MongoDB document structure, and API flow.

CODE:

Service.js:

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");
const app = express();
const PORT = 3000;
// Middleware
app.use(cors());
app.use(express.json());
// Connect MongoDB
mongoose.connect("mongodb://127.0.0.1:27017/CollegeDB")
  .then(() => {console.log("MongoDB Connected Successfully"); })
  .catch((error) => {console.log("MongoDB Connection Failed");
    console.log(error);});
// Home Route
app.get("/", (req, res) => { res.send("College Canteen Pre-Order System is Running");});
// GET /food
app.get("/food", (req, res) => { const foodItems = [
  { id: 1,
    name: "Vada Pav",
    price: 20},
  {id: 2,
    name: "Samosa",
    price: 15},
```

```

    {id: 3,
      name: "Veg Sandwich",
      price: 40},
    { id: 4,
      name: "Masala Maggi",
      price: 50},
    { id: 5,
      name: "Cold Coffee",
      price: 60} ];
res.json(foodItems));});
// POST /orders
app.post("/orders", async (req, res) => {
  try {const order = req.body;
    await mongoose.connection.db
      .collection("orders")
      .insertOne(order);
    res.json({
      message: "Order placed successfully",
      order: order
    });}
  catch (error) {
    console.log(error);
    res.status(500).json({
      message: "Error placing order"}}); });});
// GET /orders
app.get("/orders", async (req, res) => {
  try {
    const orders = await mongoose.connection.db
      .collection("orders")
      .find()
      .toArray();
    res.json(orders);}
  catch (error) {
    console.log(error);
    res.status(500).json({
      message: "Error fetching orders"}});});});
// Start Server
app.listen(PORT, () => {
  console.log(`Server is running on http://localhost:${PORT}`);});

```

App.jsx:

```

import { useEffect, useState } from "react";
import axios from "axios";
function App() {
  const [foodItems, setFoodItems] = useState([]);
  const [orders, setOrders] = useState([]);
  const [name, setName] = useState("");
  const [rollNo, setRollNo] = useState("");
  const [course, setCourse] = useState("");
  const [food, setFood] = useState("");
  const [quantity, setQuantity] = useState(1);
  // Get Food Items
  const getFood = () => {
    axios
      .get("http://localhost:3000/food")
      .then((response) => {
        setFoodItems(response.data);})
      .catch((error) => {
        console.log(error); }); });
  // Get Orders
  const getOrders = () => {
    axios
      .get("http://localhost:3000/orders")

```

```

        .then((response) => {
            setOrders(response.data);})
        .catch((error) => {
            console.log(error); }); });
// Load Food and Orders
useEffect(() => {
    getFood();
    getOrders();}, []);
// Place Order
const placeOrder = () => {
    if (name === "" ||
        rollNo === "" ||
        course === "" ||
        food === "") {
        alert("Please fill all details");
        return;}
    const selectedFood = foodItems.find(
        (item) => item.name === food);
    const newOrder = { name: name,
        rollNo: rollNo,
        course: course,
        food: selectedFood.name,
        quantity: Number(quantity),
        price: selectedFood.price,
        total: selectedFood.price * Number(quantity)};
    axios
        .post("http://localhost:3000/orders", newOrder)
        .then((response) => {
            alert(response.data.message);
            setName("");
            setRollNo("");
            setCourse("");
            setFood("");
            setQuantity(1);
            getOrders();})
        .catch((error) => {
            console.log(error);}); });
return ( <div>
    <h1>College Canteen Pre-Order System</h1>
    <hr />
    <h2>Available Food Items</h2>
    <table border="1" cellPadding="10">
        <thead>
            <tr>
                <th>Food Item</th>
                <th>Price</th>
            </tr>
        </thead>
        <tbody>
            {foodItems.map((item) => (
                <tr key={item.id}>
                    <td>{item.name}</td>
                    <td>₹ {item.price}</td>
                </tr>
            ))}
        </tbody>
    </table>
    <hr />
    <h2>Student Order Form</h2>
    <input
        type="text"
        placeholder="Student Name"

```

```

      value={name}
      onChange={(e) => setName(e.target.value)} />
    <br />
    <br />
    <input
      type="text"
      placeholder="Roll Number"
      value={rollNo}
      onChange={(e) => setRollNo(e.target.value)}
    />
    <br />
    <br />
    <input
      type="text"
      placeholder="Course"
      value={course}
      onChange={(e) => setCourse(e.target.value)} />
    <br />
    <br />
    <select
      value={food}
      onChange={(e) => setFood(e.target.value)} >
      <option value="">
        Select Food Item
      </option>
      {foodItems.map((item) => (
        <option
          key={item.id}
          value={item.name}>
            {item.name} - ₹ {item.price}
          </option>
        ))}
    </select>
    <br />
    <br />
    <input
      type="number"
      min="1"
      value={quantity}
      onChange={(e) => setQuantity(e.target.value)} />
    <br />
    <br />
    <button onClick={placeOrder}>
      Place Order
    </button>
    <hr />
    <h2>Placed Orders</h2>
    <table border="1" cellPadding="10">
      <thead>
        <tr>
          <th>Name</th>
          <th>Roll No</th>
          <th>Course</th>
          <th>Food</th>
          <th>Quantity</th>
          <th>Price</th>
          <th>Total</th>
        </tr>
      </thead>
      <tbody>
        {orders.map((order) => (
          <tr key={order._id}>
            <td>{order.name}</td>

```

```

        <td>{order.rollNo}</td>
        <td>{order.course}</td>
        <td>{order.food}</td>
        <td>{order.quantity}</td>
        <td>₹{order.price}</td>
        <td>₹{order.total}</td>
    </tr>)))}
</tbody>
</table>
</div>);}

```

export default App;

OUTPUT:

```

PS C:\Users\priti yadav\OneDrive\Desktop\Mern Practical> cd Exprac
PS C:\Users\priti yadav\OneDrive\Desktop\Mern Practical\Exprac> cd my-react
PS C:\Users\priti yadav\OneDrive\Desktop\Mern Practical\Exprac\my-react> npm install express mongoose cors

```

up to date, audited 248 packages in 12s

63 packages are looking for funding
run `npm fund` for details

found 0 vulnerabilities

```
PS C:\Users\priti yadav\OneDrive\Desktop\Mern Practical\Exprac\my-react> npm install axios
```

up to date, audited 248 packages in 2s

63 packages are looking for funding
run `npm fund` for details

found 0 vulnerabilities

```
PS C:\Users\priti yadav\OneDrive\Desktop\Mern Practical\Exprac\my-react> npm run dev
```

```
> my-react@0.0.0 dev
> vite
```

VITE v8.3.0 ready in 1067 ms

```

→ Local:   http://localhost:5173/
→ Network: use --host to expose
→ press h + enter to show help

```

```
o
o
```

```

PS C:\Users\priti yadav\OneDrive\Desktop\Mern Practical\Exprac\my-react> cd..
PS C:\Users\priti yadav\OneDrive\Desktop\Mern Practical\Exprac> node service.js
Server is running on http://localhost:3000
MongoDB Connected Successfully

```

← → ↺ ⓘ localhost:3000/food

Pretty print ☒

```

[
  {
    "id": 1,
    "name": "Vada Pav",
    "price": 20
  },
  {
    "id": 2,
    "name": "Samosa",
    "price": 15
  },
  {
    "id": 3,
    "name": "Veg Sandwich",
    "price": 40
  },
  {
    "id": 4,
    "name": "Masala Maggi",
    "price": 50
  },
  {
    "id": 5,
    "name": "Cold Coffee",
    "price": 60
  }
]

```

College Canteen Pre-Order System

Available Food Items

Food Item	Price
Vada Pav	₹20
Samosa	₹15
Veg Sandwich	₹40
Masala Maggi	₹50
Cold Coffee	₹60

Student Order Form

Student Name

Roll Number

Course

Select Food Item

1

Place Order

Placed Orders

Name	Roll No	Course	Food	Quantity	Price	Total
------	---------	--------	------	----------	-------	-------

College Canteen Pre-Order System

Available Food Items

Food Item	Price
Vada Pav	₹20
Samosa	₹15
Veg Sandwich	₹40
Masala Maggi	₹50
Cold Coffee	₹60

Student Order Form

Priti Yadav

T054

IT

Veg Sandwich - ₹40

1

Place Order

Placed Orders

Name	Roll No	Course	Food	Quantity	Price	Total
------	---------	--------	------	----------	-------	-------

College Canteen Pre-Order System

Food Item	Price
Vada Pav	₹20
Samosa	₹15
Veg Sandwich	₹40
Masala Maggi	₹50
Cold Coffee	₹60

localhost:5173 says
Order placed successfully

OK

Student Order Form

Priti Yadav

T054

IT

Veg Sandwich - ₹40

1

Place Order

Placed Orders

Name	Roll No	Course	Food	Quantity	Price	Total
------	---------	--------	------	----------	-------	-------

College Canteen Pre-Order System

Available Food Items

Food Item	Price
Vada Pav	₹20
Samosa	₹15
Veg Sandwich	₹40
Masala Maggi	₹50
Cold Coffee	₹60

Student Order Form

Student Name

Roll Number

Course

Select Food Item

1

Place Order

Placed Orders

Name	Roll No	Course	Food	Quantity	Price	Total
Priti Yadav	T054	IT	Veg Sandwich	1	₹40	₹40

CONCLUSION:

The College Canteen Pre-Order System is a web-based application developed using React, Axios, Node.js, Express.js, and MongoDB.

It allows students to view available food items and their prices.

Students can enter their details, select a food item, and specify the quantity.

The order is sent to the server using a POST API and stored in MongoDB.

The GET API retrieves the placed orders and displays them on the React interface.

Thus, the project provides a simple and efficient way to pre-order food from the college canteen.